

MATH 410 - WINTER 2026 - HOMEWORK 5

1. Let $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Compute e^{tJ} (where $t \in \mathbb{R}$).

2. Recall the Bloch equation for the magnetization vector $M : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in the external magnetic field $B = B(t, x, y, z) \in \mathbb{R}^3$:

$$\frac{d}{dt}M = \gamma[M \times B] - \frac{1}{T_2}M^\perp - \frac{1}{T_1}[M^\parallel - M_0],$$

where T_1, T_2 are the relaxation constants, $M^\perp$ is the transverse component of M relative to B , and $M^\parallel$ is the longitudinal component of M relative to B .

Let $b_0 \in \mathbb{R}$, $\omega_0 := \gamma b_0$, and define

$$U(t) = \begin{bmatrix} \cos(\omega_0 t) & \sin(\omega_0 t) & 0 \\ -\sin(\omega_0 t) & \cos(\omega_0 t) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Define $m = m(t, x, y, z)$ via

$$M(t, x, y, z) = U(t)m(t, x, y, z).$$

Show that m satisfies the equation

$$\frac{d}{dt}m = \gamma[m \times B_{\text{eff}}] - \frac{1}{T_2}m^\perp - \frac{1}{T_1}[m^\parallel - M_0],$$

where

$$B_{\text{eff}} = U^{-1}(t)B - (0, 0, b_0)$$

and $m^\perp, m^\parallel$ are defined analogously.

Suggestion. First set $T_1 = T_2 = \infty$ (i.e. ignore the relaxation terms) and do the calculation. Then check the relaxation terms once you've got the hang of it.

3. Show that if A is skew-symmetric (i.e. $A^* = -A$) then e^A is unitary (i.e. $(e^A)^*(e^A) = I$).

4. Identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(z_1, z_2) \leftrightarrow z_1 + iz_2$. Let $z = z_1 + iz_2$. Show that the linear transformation $T_z : \mathbb{C} \rightarrow \mathbb{C}$ given by $T_z(w) = zw$ is represented (as a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$) by the matrix

$$\begin{bmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{bmatrix}.$$

5. Suppose $B \in \mathbb{R}^3$ and $M = M(t) \in \mathbb{R}^3$ satisfies

$$\frac{d}{dt}M = M \times B.$$

Show that $|M(t)|^2$ is constant.